PART A – SADC BRANCH RESEARCH

Johannesburg, South Africa – Kalahari Logistics Ltd

It focuses on the Johannesburg SADC partner office and connects the research to the network design already implemented in Cisco Packet Tracer.

A1. Introduction to the SADC Branch

Kalahari Logistics Ltd includes a Johannesburg branch in South Africa as its SADC partner office. In the final network design, Johannesburg is connected to the Windhoek headquarters through the central Windhoek router. The Johannesburg LAN uses the private IPv4 subnet 192.168.40.0/24, with JHB-RTR using 192.168.40.1 as the default gateway, JHB-SW using 192.168.40.2 for management, and two branch PCs using 192.168.40.10 and 192.168.40.11. The WAN connection between Windhoek and Johannesburg uses the /30 network 10.0.0.4/30, with 10.0.0.5 on WH-RTR and 10.0.0.6 on JHB-RTR. This gives the SADC branch a clearly separated LAN while allowing communication with the other company sites.

A2. Wireless and Mobile Network Infrastructure in South Africa

South Africa has a mature telecommunications market with extensive mobile and fixed broadband infrastructure. According to ICASA's 2025 State of the ICT Sector report, national population coverage reached 99.07% for 4G/LTE and 99.79% for 3G, while 5G population coverage was 46.64%. The same report indicates that broadband geographical coverage had reached 82.06% of the country's landmass. These figures show that mobile broadband is an important connectivity option for businesses, including logistics organisations that need connectivity beyond a fixed office environment.

Johannesburg is one of South Africa's major commercial centres, so a business branch can make use of several connectivity technologies. 4G/LTE provides a practical backup or alternative to fixed broadband, while 5G can provide higher capacity and lower latency where coverage is available. Vodacom, MTN, Telkom and other licensed operators contribute to the mobile and fixed connectivity environment. For a logistics company, mobile connectivity can support staff mobility, communication, tracking systems and business continuity when a fixed connection is unavailable.

A3. Fibre, WAN and Business Connectivity

Fibre is important for a branch office because it can provide a high-capacity and stable connection for business applications, cloud services, voice/video communication and inter-site traffic. South Africa has a large fibre ecosystem involving network operators and fibre network providers. Vodacom, for example, lists several third-party network providers through which its fibre broadband services may be delivered, including Openserve, Vumatel, Frogfoot, MetroFibre Networx, Link Africa, MTN and others.

For Kalahari Logistics, the Johannesburg branch can use a business-grade fixed connection as its primary WAN access while the simulated serial WAN link represents the logical connection to Windhoek. In the Packet Tracer design, JHB-RTR performs the routing function for the local

Johannesburg LAN and forwards traffic towards WH-RTR. This reflects the principle that the branch LAN and the inter-site WAN should be logically separated and managed through the branch router.

A4. International and Regional Connectivity

South Africa also has strong international connectivity through submarine cable systems. Current industry information lists international subsea cables landing in South Africa, including WACS, SEACOM, EASSy, ACE, Equiano and 2Africa, among others. These systems provide international capacity and contribute to South Africa's role as a major regional connectivity hub.

This international connectivity is relevant to Kalahari Logistics because the project connects Windhoek, Johannesburg and London. The Johannesburg office is therefore not only a SADC connection point but also part of a wider regional and international network. In a real deployment, traffic between Namibia, South Africa and overseas services could use carrier networks, fibre backbones and international submarine cable systems. The Packet Tracer model simplifies these real-world carrier networks into direct router-to-router WAN links so that routing and connectivity can be demonstrated clearly.

A5. Regulation and Cross-Border Connectivity

The Independent Communications Authority of South Africa (ICASA) is the country's official regulator for telecommunications, broadcasting and postal services. Its responsibilities include licensing electronic communications services and networks, managing radio-frequency spectrum, monitoring compliance and protecting consumers. ICASA operates under the ICASA Act and the Electronic Communications Act.

For a regional business such as Kalahari Logistics, regulation matters because connectivity between Namibia and South Africa involves different national regulatory environments, licensed service providers and cross-border infrastructure. Regional cooperation and compatible technical and regulatory approaches can make it easier for businesses to maintain reliable communications across SADC countries. ICASA also identifies international and regional cooperation as part of its mandate.

A6. Comparison with the Namibian Headquarters

Area	Johannesburg SADC Branch	Windhoek HQ
Mobile connectivity	Strong 4G/LTE coverage and expanding 5G coverage nationally.	Mobile connectivity is important for urban and rural users, with 4G/5G increasingly supporting business mobility.
Fixed connectivity	Large fibre and business broadband ecosystem.	Fibre and other WAN options can support the HQ, while alternatives may be important where fibre is unavailable.
Regional role	Major SADC commercial and connectivity hub.	Central HQ connecting the company's Namibia and international sites.

Project role	JHB-RTR connects the Johannesburg LAN to WH-RTR.	WH-RTR acts as the central router connecting Johannesburg, Walvis Bay and London.

The comparison shows why Johannesburg is useful as the selected SADC branch in the project. South Africa provides extensive telecommunications infrastructure and is geographically and economically important for regional logistics. Namibia remains the company's headquarters and central control point, while Johannesburg provides a strong regional connection into the wider SADC environment.

A7. Application to the Kalahari Logistics Network

The research supports the technical design used in the final Packet Tracer project. Johannesburg is represented by
JHB-RTR, JHB-SW and two PCs. JHB-RTR uses 192.168.40.1/24 on its LAN interface and 10.0.0.6/30 on the WAN link to
Windhoek. The branch switch is managed using 192.168.40.2/24 with a default gateway of 192.168.40.1. The two PCs use
192.168.40.10 and 192.168.40.11. The final report records successful cross-site testing, including a ping from Windhoek to 192.168.40.11 and return connectivity from a Walvis Bay PC to a Windhoek host.

These tests demonstrate the key purpose of the SADC branch: users in Johannesburg must be able to communicate with the headquarters and other company sites through the WAN. The design also keeps the Johannesburg LAN on its own subnet, reducing addressing overlap and making the network easier to manage.

A8. Recommendations for the SADC Branch

First, the Johannesburg office should use a reliable business-grade fibre or equivalent fixed connection as its primary service where available. Second, 4G/5G should be considered as a backup link to improve business continuity. Third, the organisation should use secure routing and appropriate firewall controls between the branch and HQ. Fourth, the branch should use network monitoring so that WAN latency, packet loss and availability can be identified quickly. Finally, as the organisation grows, a dynamic routing protocol such as OSPF could replace the static routing approach used in the simulation, improving scalability. VLAN segmentation could also be introduced if the Johannesburg office adds departments or more users.

A9. Conclusion

Johannesburg is an appropriate SADC branch for Kalahari Logistics because it provides access to South Africa's extensive mobile, fibre, regional and international connectivity infrastructure. The branch can support logistics operations through reliable fixed connectivity, mobile backup options and a routed WAN connection to Windhoek. The research also supports the project's Packet Tracer design, where JHB-RTR provides the gateway for the 192.168.40.0/24 LAN and connects the branch to the central Windhoek router. Together, the research and simulation demonstrate how a logistics company can design a multi-site network that supports regional communication and future expansion.

References

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